Mammalia Raccoon Proximity Network Analysis

https://github.com/aiyshwary/Raccoon-Network-Analysis

Python NetworkX Jupyter Graph Analysis Epidemiology

OVERVIEW

A graph-based epidemiological study modelling rabies transmission risk in urban raccoon populations through proximity network analysis and centrality measures, supporting public health and wildlife management decisions.

IMPACT

Constructed and analyzed a raccoon proximity network using real-world contact data, applying centrality measures to identify high-risk individuals and quantify potential rabies spread pathways within urban populations.

TECHNICAL WALKTHROUGH

Mammalia Raccoon Proximity Network Analysis – Project

Explanation

Project Overview

This project analyzes social interaction patterns among urban raccoons using network analysis to understand how rabies and other pathogens can spread within raccoon populations.
Since raccoons are a major vector of rabies, studying how often and how closely they interact helps estimate disease transmission risk and supports better public health and wildlife management strategies.

Objective

Model raccoon interactions as a graph/network

Identify high-risk raccoons that can spread pathogens quickly Use centrality measures to understand social structure and disease
flow

Dataset & Network Construction

24 raccoons → represented as nodes
~2000 interactions → represented as weighted edges

Data collected using proximity-detecting collars

Edge weight = time spent in close proximity Monthly interaction data used to build the network If two raccoons are connected, it means they had close physical interaction, which is critical for rabies transmission.
Why Network Analysis?
Disease spread is not random.

It depends on

i) Who interacts with whom

ii) How frequently

iii) Who acts as a bridge between groups

iv) Network analysis helps identify super-spreaders and critical connectors.

Centrality Measures Used & Interpretation

Degree Centrality

i) Measures number of direct connections

ii) High degree = more interactions

Inference

i) Raccoon #4 has the highest degree

ii) Acts as a high-contact individual

iii) High risk of initiating widespread infection

Closeness Centrality
Measures how close a node is to all others

Indicates how fast something can spread
Raccoon #2 has highest closeness centrality
Can spread pathogens rapidly across the entire network

Betweenness Centrality
Measures how often a node lies on shortest paths

Identifies bridge nodes
Raccoon #16 connects different subgroups

Acts as a key transmission bridge
Removing or vaccinating this raccoon can significantly slow disease spread

Clustering Coefficient
Measures how tightly a raccoon is embedded in a local group

Indicates family or group behavior
Raccoon #2 has highest clustering
Infection here may cause localized outbreaks within families

Eigenvector Centrality
Measures influence based on connections to influential nodes
Raccoon #4 again ranks highest

Connected to other highly interactive raccoons

Considered a super-spreader candidate

PageRank Algorithm
Evaluates overall importance considering direction and weight of interactions

Helps rank raccoons by global influence

Key Findings

Raccoon populations are highly interconnected

Disease can spread much faster than expected
A few raccoons play disproportionately large roles in pathogen transmission
Targeted intervention (vaccination/tracking of key raccoons) is more effective than random control

Technologies & Concepts Used

Graph Theory

Weighted graphs

Centrality algorithms

Epidemiological modeling concepts

KEY DETAILS

1. Built a weighted proximity network from the mammalia-raccoon-proximity dataset using Python and NetworkX.

2. Computed degree, betweenness, closeness, and eigenvector centrality to rank individuals by transmission risk.

3. Identified superspreader nodes whose removal would most disrupt disease propagation chains.

4. Visualized network topology and centrality distributions to communicate findings clearly.

5. Results provide evidence-based recommendations for targeted wildlife intervention strategies.

6. Analyzed a dense 24-node, 226-edge proximity graph (~82% density) derived from 1,997 weighted temporal contact records.

7. Computed clustering coefficients and aggregated per-node contact strength (total edge weight) to quantify interaction intensity — node 4 had the highest total contact weight (956,341).

8. Rendered the network using Kamada-Kawai force-directed layout (Matplotlib) to reveal spatial clustering patterns in the proximity data.